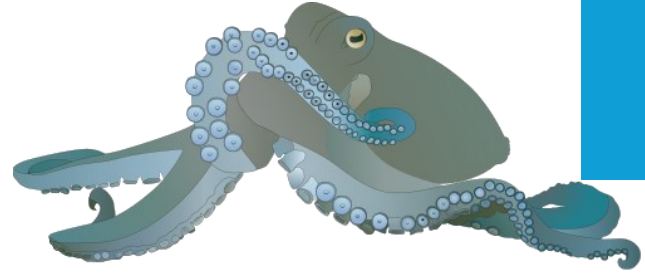




Project Proposal

Octopus Under Threat:

A Co-Managed Approach to Reduce Illegal,
Unreported and Unregulated Fishing (IUU)
in the Algarve, Portugal



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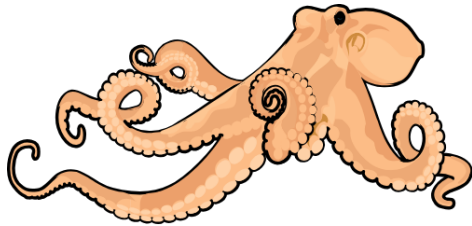
Topics in Aquaculture, Fisheries and Conservation – 14301043

Leonard Lutter (Student No.: 90017),
Eleonora Savino (Student No.: 89606),
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Overview

Octopus

- Common octopus (*Octopus vulgaris*)
- National total landings: 6,244 ton (48,5 M €)
- Among most profitable species in PT fisheries (at > 7€/kg)



(IAN, 2024)

- Highly susceptible to environmental impacts
- Terminal breeders; implications for fisheries
- long breeding season variable growth

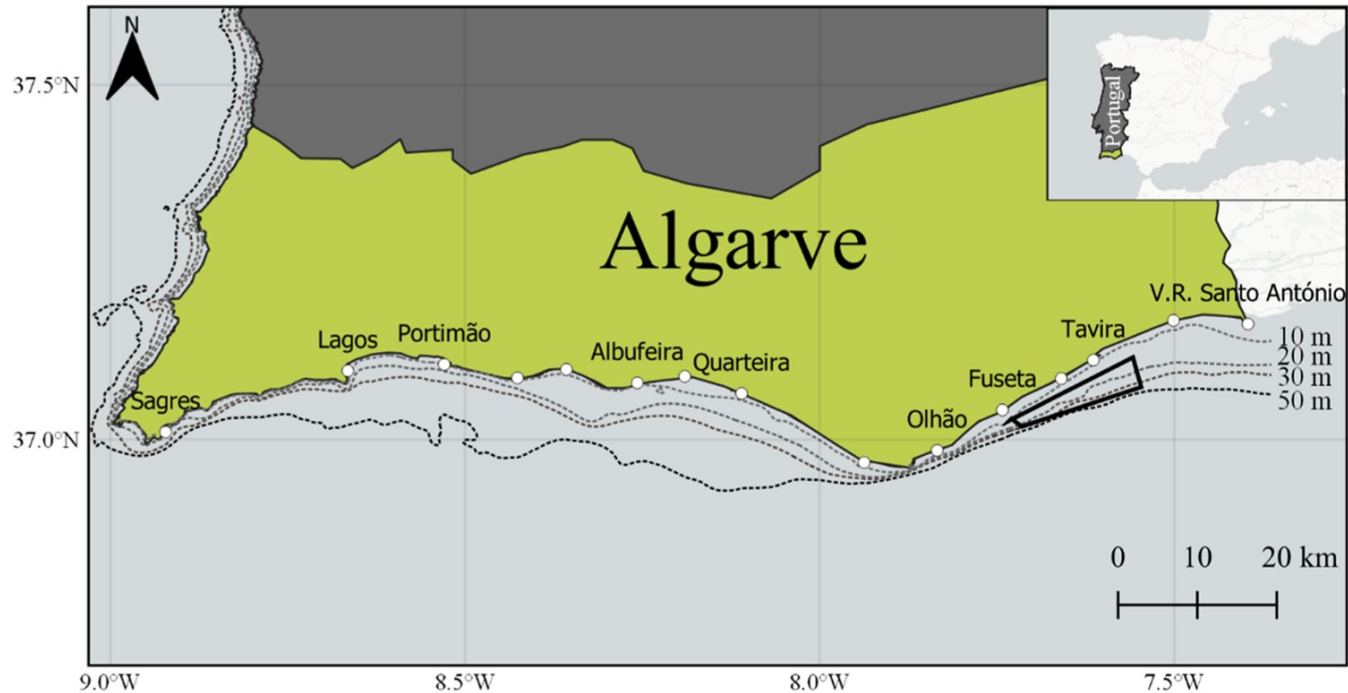
(INE, 2024; Pita et al., 2015)

Overview

Algarve

- Most important region for pot & trap Octopus fishery in Portugal
- Accounts for 52% of the national landings
- 90% octopus catch from pots & trap fishery
- High socio-economic importance (700+ people employed)
- 358 small-scale fishing vessels among 14 fisheries associations

(ParticiPesca, 2024; Pita et al., 2015; Silva, 2019; Sonderblohm, 2017)



Algarve region and cities with main fishing ports (Leitao et al., 2014)

Overview

Management

- Excluded from quota regulations under the Common Fisheries Policy (CFP)
- Previously characterized by a top-down decision-making process, with sporadic participation of fishers
- Min. landing weight (750g) & gear restrictions are main control measures

Low level of compliance with rules and regulations !

(Gaspar et al., 2014; Pita et al., 2015)

Overview

Main Concerns (Fishers)

- Illegal fishing practices; lack of enforcement
(**Gaspar et al., 2014**)
- Illegal selling from recreational fisherman / IUU fisheries (**Pascual-Fernández et al., 2020**)



ParticiPESCA



(participesca.pt, 2024)

Project

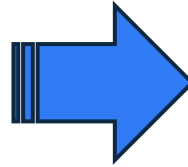
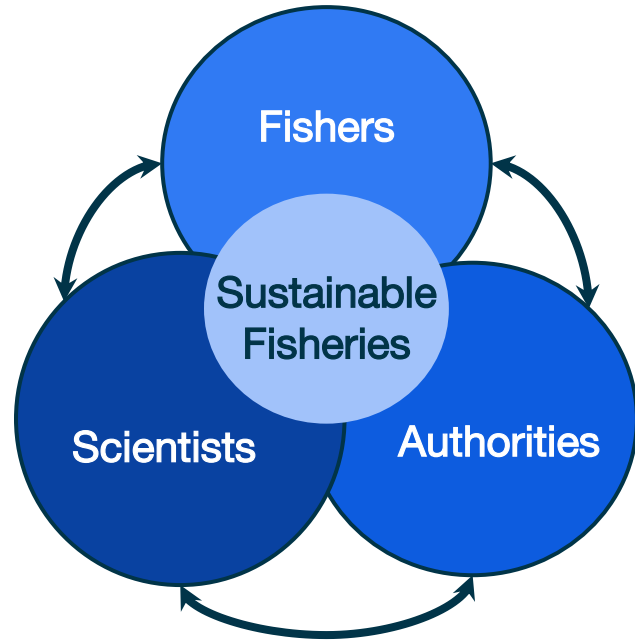
- Sustainable resource management for the pot and trap Octopus fishery in the Algarve

Objectives

- Implementing a Co-Management Committee (03/24)
- Proposal of Co-Management-Plan until end of 2024

(ParticiPesca, 2024)

Co-Management



**Empower
fishers and
Collaboration**



**Enhanced
Monitoring**



**Ensure
Sustainability**

Illegal, Undocumented and Unregulated Fishing (IUU)

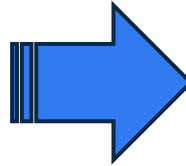
IUU Fishing refers to **Illegal, Unreported, and Unregulated fishing**, which includes activities that:

- **Illegal:** Violate national or international fishing laws.
- **Unreported:** Are not declared or misreported to authorities.
- **Unregulated:** Occur in areas without conservation measures or disregard them.

It threatens sustainable fisheries, marine biodiversity, food security, and livelihoods.

FAO, 2024

Illegal, Undocumented and Unregulated Fishing (IUU)



Did you know that 1/5 of
octopus landings in
Portugal go unreported?

Causes?

- Low profits
- High costs
- Weak enforcement of fishing rules
- Limited patrols
- Minimal compliance

Research Proposal

- Research aim: Fill critical gaps in understanding IUU fishing in the Algarve's octopus fishery
- Main objectives
 - Identifying underlying reasons for IUU fishing
 - Developing solutions within the co-management committee
 - Investigating where IUU octopus is sold and finding ways to prevent illegal sales
- Improve the sustainability of the octopus fishery and reduce IUU practices

Research design

Collaboration with the Algarve octopus fishing co-management committee

1. Systematic Literature Review

- Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (Page et al. 2021)
- foundation for project
- review of peer-reviewed and gray literature
- focus on co-management case studies and IUU fishing policy documents

Research design

Collaboration with the Algarve octopus fishing co-management committee

May
2025

1. Systematic **literature review** on co-management and IUU fishing

2. Interviews

- anonymous, semi-structured qualitative interviews (Flick 2007)
- 25-30 members of the fishing community (fishers, members from NGOs, scientists, supply chain actors)
- snowball sampling to identify further interviewees (Naderifar et al., 2017)
- key themes: perceptions, reasons, observations & possible solutions for IUU fishing
- subsequent qualitative content analysis (based on Mayring 2010) using MAXQDA (VERBI Software, 2021)

Research design

Collaboration with the Algarve octopus fishing co-management committee

May
2025

1. Systematic **literature review** on co-management and IUU fishing



2. Interviews

- semi-structured qualitative interviews → perceptions of IUU fishing & possible solutions
- qualitative content analysis using MAXQDA

October -
November
2025

3. Workshop

- invite stakeholders
- presentation of the intermediate results
- focus group discussions in smaller groups and plenary (ONyumba et al., 2018)
- vision of a co-managed octopus fishery without IUU fishing
- backcasting process to develop interventions to reach vision (Baumann et al., 2023)

Research design

Collaboration with the Algarve octopus fishing co-management committee

1. systematic **literature review** on co-management and IUU fishing

May
2025



2. Interviews

- semi-structured qualitative interviews → perceptions of IUU fishing & possible solutions
- qualitative content analysis using MAXQDA

October -
November
2025



3. Workshop

- intermediate results presentation
- focus group discussions
- visioning: co-managed octopus fishery without IUU fishing

Spring
2026

4. Final report

- Summary of results from literature review, interviews, focus group discussion & visioning exercise
- present to co-management committee
- including policy brief
- foundations for further steps towards integrating the ideas for the prevention of IUU fisheries and sale

Research design

Collaboration with the Algarve octopus fishing co-management committee

1. systematic **literature review** on co-management and IUU fishing

May
2025

2. **Interviews**

- semi-structured qualitative interviews → perceptions of IUU fishing & possible solutions
- qualitative content analysis using MAXQDA

October -
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2025

3. **Workshop**

- intermediate results presentation
- focus group discussions
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Spring
2026

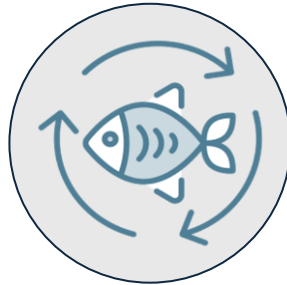
4. **Final report**, presentation & policy brief

January
2027

Conclusion: Why fund this Project?

Urgency:

Safeguard marine ecosystems
and livelihood of fishermen



Our Solution:

A co-management framework
uniting fishers, scientists, and
policymakers

Impact:

Immediate and long-term
benefits



Value:

Investment in sustainable
fisheries with measurable
outcomes

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Thank you for listening!
Questions?

